Evaluation of the role of O-GlcNAcylation in liver regeneration – Differential gene expression and pathway analysis of OGA-inhibited mice liver samples

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Abstract: O-GlcNAcylation is a type of post-translational modification where a single N-acetylglucosamine (GlcNAc) sugar dynamically cycles on and off serine and threonine residues of proteins by the action of O-GlcNAc Transferase (OGT) and O-GlcNAcase (OGA) respectively. Disrupted O-GlcNAc homeostasis affects various cellular functions and disease progression. Of note, decreased O-GlcNAcylation was found to disrupt liver regeneration by promoting hepatocyte proliferation, eventually forming liver tumors. This phenotype was reversed in liver specific OGA KO mice. Thus, in this study we increased O-GlcNAcylation via Thiamet-G (OGA inhibitor) treatment to assess if it helps to promote liver regeneration. To identify the genes that could be involved in this process we used RNA-Seq analysis. However, out of 12775 genes analyzed, only 19 genes were downregulated, 10 genes were upregulated, and 12,746 genes had no significant changes in expression. Some of the top significantly regulated genes included Plin4, SELENBP2, MANF and OGA. Gene ontology (GO) pathway analysis showed some significantly enriched pathways including intracellular anatomical structure, cytoplasm and cellular anatomical entity. Interestingly, Gene Set Enrichment Analysis (GSEA) showed Hepatic graft versus host disease (GVHD) was downregulated with TMG treatment. However, we did not find more specific pathways related to liver regeneration. Furthermore, hierarchical clustering and k-means clustering analysis to explore if the genes display any clustering pattern revealed no specific clustering of genes. Heatmap also showed that there was no clustering of genes between the saline and TMG groups. Hence, we conclude that the RNA-Seq analysis must be repeated with longer duration of TMG injection (at least one month or up to six months TMG injection) and with liver-specific genetic OGT knockout (KO) and OGA KO mice for effective comparison. Repeating the RNA-Seq analysis with these changes will lead to the identification of significant changes in gene expression due to O-GlcNAc disruptions in mice liver, thus helping to understand the mechanism by which O-GlcNAcylation regulates liver regeneration.

Keywords: O-GlcNAc, OGT, OGA, TMG, mice-liver, RNA-Seq, pathway analysis

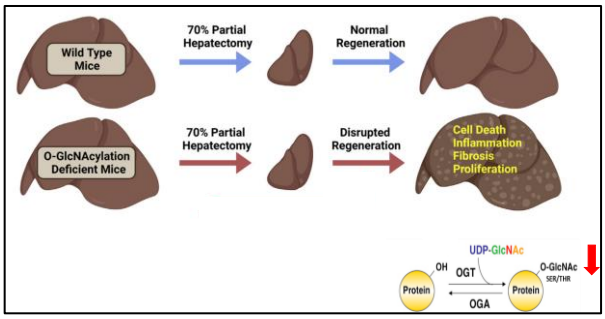

Figure 1: O-GlcNAc deficient mice have disrupted liver regeneration along with cell death, inflammation, fibrosis and proliferation.

Introduction: O-GlcNAcylation is a type of post-translational modification that involves the dynamic addition and removal of a single N-acetylglucosamine (GlcNAc) sugar to serine and threonine residues of proteins. The addition of O-GlcNAc is catalyzed by the enzyme O-GlcNAc transferase (OGT) and the removal is catalyzed by O-GlcNAcase (OGA) (1). UDP-GlcNAc is the substrate for OGT from which OGT takes GlcNAc and adds to serine/ threonine residues of proteins to form O-GlcNAcylated proteins. Multiple metabolic pathways feed into the Hexosamine Biosynthetic Pathway (HBP) to generate UDP-GlcNAc making O-GlcNAcylation a nutrient sensor. Thus, O-GlcNAc dynamically cycles on and off proteins by the action of OGT and OGA respectively, referred to as O-GlcNAc cycling (2). However, O-GlcNAc quickly returns to a base-line level after stimulus removal, termed O GlcNAc homeostasis (1-4). O-GlcNAcylation plays a critical role in various biological processes including cell proliferation, differentiation, and disease progression (5, 6). It was recently shown that O-GlcNAcylation in hepatocytes is essential for liver regeneration. Without O-GlcNAcylation, hepatocytes keep on proliferating,

eventually forming liver tumors (7) **(Figure 1)**. Thiamet-G (TMG) is a drug that inhibits OGA and hence increases O-GlcNAcylation (4). Hence, increased O-GlcNAcylation due to TMG treatment could aid liver regeneration. To further understand the role of O-GlcNAc mediated regulation of liver regeneration and the genes involved in this process, we did a pilot study of RNA Sequencing (RNA-Seq) Analysis with mice injected for two weeks with TMG (4, 7).

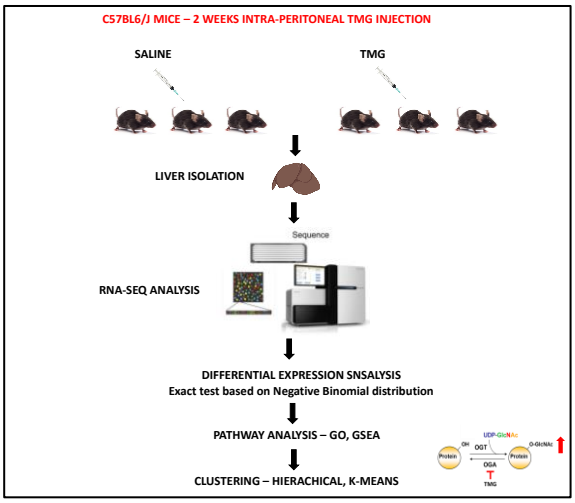

Figure 2: Experimental Outine for RNA-Seq analysis: RNA was extracted from the livers of control and TMG injected mice. RNA-Seq analysis was done using EdgeR – exact test to identify differentially expressed genes. Pathway analysis using gene ontology (GO) and Gene Set Enrichment Analysis (GSEA) was done. Gene clustering analysis was also performed using hierarchical and k-means clustering.

Briefly three mice were injected intraperitoneally for two weeks with TMG and three mice were injected with saline for control (8). The mice were euthanized after two weeks, and the livers were isolated. RNA was extracted from the livers and RNA sequencing analysis was done (7). The genes that were differentially expressed were analyzed using EdgeR via exact test and pathway analysis was done using gene ontology (GO) and gene set enrichment analysis (GSEA) (4, 7). Gene cluster analysis was also done to see if the genes exhibit any clustering pattern and heatmap was plotted. Overall, the results indicated that only a few genes were regulated significantly with two weeks of TMG injection. Thus, further analysis is needed with longer duration of injection (at least one month or up to six months TMG injection as observed with wet-lab analysis for other projects) and with liver-specific genetic OGT knockout (KO) and OGA KO mice (7, 9).

Materials and methods:

Animal protocols and models: The University of Kansas Medical Center Animal Care and Use Committee approved all experiments in this study. Two-month-old male C57BL/6J mice were purchased from The Jackson Laboratory (Bar Harbor, ME). All mice were housed using a standard 12-h light/dark cycle with access to chow and water ad libitum. Mice were treated with a 50 mg/kg TMG intra-peritoneal injection every other morning for 2 weeks (10).

Total RNA isolation: Total RNA was isolated by using TRI Reagent solution (Ambion) per the manufacturer's instruction. 1 ml of TRI Reagent solution was added to each brain sample, and then 200 ul of chloroform (Fisher) was added to extract RNA. Upon centrifugation, an equal amount of isopropyl alcohol (Fisher) was added to the upper layer solution containing RNA. RNA pellets were precipitated by centrifugation, washed once with 70% ethanol, and air-dried, and the pellet was dissolved in nuclease-free water (Life Technologies, Inc.) (14). RNA concentration was measured using a Nanodrop 2000c (Thermo Fisher Scientific) (4)

cDNA library preparation: cDNA library was prepared using Illumina TruSeq Stranded mRNA sample preparation kit (Illumina) as per the manufacturer's instruction. 800 ng of the total RNA per reaction was used to initiate the protocol (4).

RNA sequencing analysis: The quality of RNA sequencing results was first assessed using FastQC (0.11.2). RSEM (1.2.22) was utilized to align the reads to the human reference genome HG38 and to calculate gene expression values. Differential expression analysis was carried out in R programming language using EdgeR (3.14.0) as there are only two groups – Saline and TMG injection (4). First, genes with gene counts as zero for

all samples were removed. The quality of the samples was checked by MDS (Multidimensional Scaling) plot, Mean Variance plot by gene, Barplot, Boxplot (In log10 scale) and Density plots by sample. Only genes with counts per million (cpm) greater than or equal to 2 were used for analysis to reduce the burden of multiple testing in differential gene expression analyses. A DGE list was created to carry out differential gene expression analysis. Normalization was carried out using the TMM-method (weighted trimmed mean of M-values), the commonly used method for differential expression analysis using EdgeR. Gene wise dispersion parameter adjusted using Bayesian empirical method was calculated. The quality of the samples after filtering was checked again by MDS plot, Mean Variance plot by gene, Barplot, Boxplot (In log10 scale) and Density plots by sample. Finally, differential gene expression was calculated using EdgeR by computing gene-wise exact tests for differences in the means between two groups of negative binomially distributed counts (11).

Gene Ontology (GO) and Gene Set Enrichment Analysis (GSEA): The Ensemble ID of differentially expressed genes ranked according to their p value were mapped to gene symbol. Gene ontology (GO) analysis was carried out to the mice (*Mus musculus*) database using the goana function in R. All gene sets file for *Mus musculus* was downloaded from Broad Institute. Gene Set Enrichment Analysis (GSEA) was carried out using fgsea function. A table of top 10 upregulated and downregulated pathways was made, and enrichment plots were plotted for the top three upregulated and downregulated pathways (11).

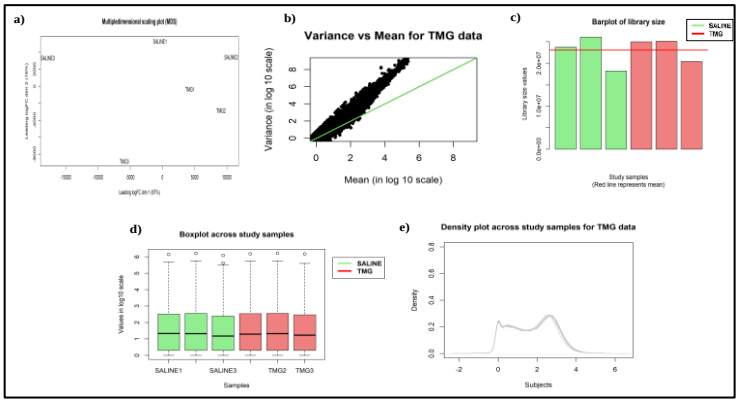

Figure 3: Plots for quality control check of RNA-Seq data: a) Multi-Dimensional scaling (MDS) -plot shows some grouping of subject 3. **b)** Mean Variance plot by gene shows there is over-dispersion of genes. **c)** Barplot shows subject 3 has lower library size **d)** Boxplot (in log10 scale) shows the median line is not perfectly aligned, indicating normalization might be necessary. **e)** Density plots by sample indicate uniform distribution of samples.

Gene Clustering Analysis: Hierarchical clustering and k-means clustering analysis was done to explore if the genes display any clustering pattern and the genes in clusters were assessed using CoxPH analysis. A heatmap of genes was also plotted to see if the genes display any common expression pattern between the control and treatment groups. The following R-packages were utilized for calculations and visualizations: ggplot2 and edgeR. The Benjamini and Hochberg procedure was used to control the false discovery rate (11, 12) **(Figure 2)**.

Results:

Quality control of RNA-Seq data: Samples had two groups - saline and TMG injected mice and each group had three subjects (three mice). First, the quality of the samples was checked by MDS (Multidimensional Scaling) plot, Mean Variance plot by gene, Bar plot, Boxplot (In log10 scale) and Density plots by sample. The MDS plot showed saline 3 and TMG 3 (3rd Subject) were grouped separately from the rest of the subjects, indicating normalization is necessary. Mean and Variance relationship plot shows variance is greater than mean and RNA Seq data is over-dispersed; over-dispersion of RNA-Seq data is necessary to carry out RNA-seq analysis as it indicates there is difference in expression between genes. The bar plot looked good, however, sample 3 seemed to have lower library size compared to the other samples. This further indicates normalization of data is necessary.

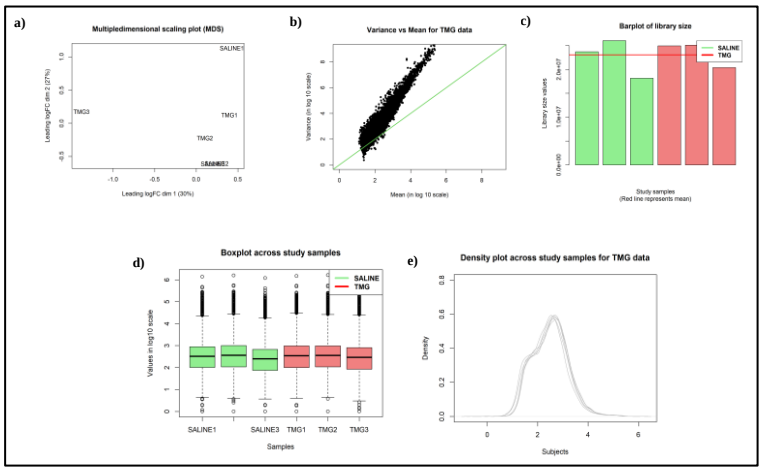

Figure 4: Quality control check plots replotted after normalization using TMM method: a) Multi-Dimensional scaling (MDS) -plot shows some grouping of TMG subject 3. **b)** Mean Variance plot by gene shows there is over-dispersion of genes. **c)** Barplot shows subject 3 has lower library size **d)** Boxplot (in log10 scale) shows the median line is aligned after normalization, except for subject 3 **e)** Density plots by sample indicate uniform

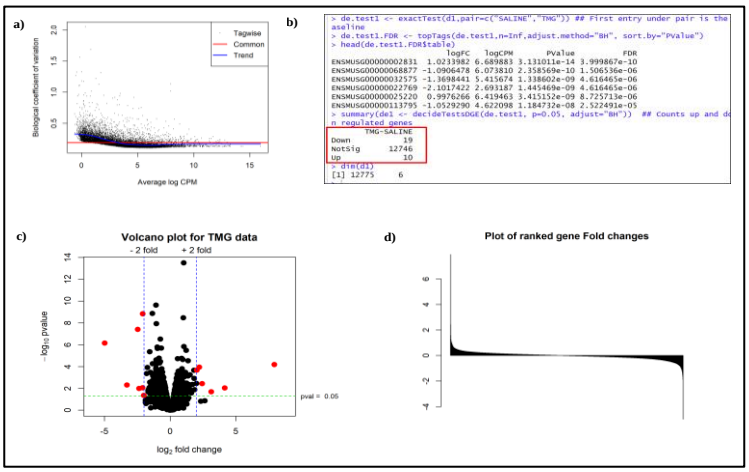

Figure 5: Differential Gene Expression Analysis: a) The Biological Co-efficient of Variation (BCV) plot shows some genes have higher varaition compared to other genes. **b)** Snapshot of significantly regulated genes show 19 genes were downregulated, 10 were upregulated and 12,746 genes had no significant change in expression. **c)** The volcano plot highlights differentially expressed genes with $p < 0.05$ and log-fold change > 2 **d)** Plot of ranked gene fold changes indicates a few genes are significantly up and down regulated.

The median line of boxplot also needs to be aligned. Density plots showed the samples have uniform distribution. However, since the MDS plot, box plot and bar plot indicate slight variation with sample 3, normalization was carried out using weighted trimmed mean of M-values (TMM) **(Figure 3)**.

Normalization of RNA-Seq data: Genes with gene counts as zero for all samples were removed. Only genes with counts per million (cpm) greater than or equal to 2 were used for analysis to reduce the burden of multiple testing in differential gene expression analyses. A DGE list was created to carry out differential gene expression analysis using EdgeR. Normalization was carried out using the TMM-method, the commonly used method for differential expression analysis using EdgeR. TMM normalization method is based on the hypothesis that most genes are not differentially expressed (DE). Gene wise dispersion parameter adjusted using Bayesian empirical method was calculated. The quality of the samples after filtering was checked again by MDS (Multidimensional Scaling) plot, Mean Variance plot by gene, Barplot, Boxplot (In log10 scale) and Density plots by sample. Even after normalization some grouping was found with TMG sample 3 and the library size of TMG sample 3 looked small **(Figure 4)**.

Differential Gene Expression Analysis: Finally, differential gene expression analysis was carried out for the filtered data using EdgeR by computing gene-wise exact tests for differences in the means between the two groups (Saline and TMG) of negative binomially distributed counts. EdgeR was used as there are only two groups – Saline and TMG injection. Out of 12775 genes, 19 were downregulated, 10 were upregulated and 12,746 genes were not significant. The Biological Co-efficient of Variation (BCV) plot and volcano plots were plotted to visualize the differential gene expression. BCV plot showed that some genes have higher Biological coefficient of Variation or greater variation compared to other genes. The genes having significant regulation were also shown by volcano

plot. The volcano plot highlights genes with a p value less than 0.05 and a log-fold change greater than two. Plot of ranked gene fold changes shows only a few genes up and down regulated **(Figure 5)**.

Pathway analysis using Gene Ontology (GO) and Gene Set Enrichment Analysis (GSEA): The Ensemble ID of differentially expressed genes ranked according to their p value were mapped to gene symbol. Gene ontology (GO) analysis was carried out to the mice (*Mus musculus*) database using the goana function in R. GO analysis showed some significantly enriched pathways including intracellular anatomical structure, cytoplasm and cellular anatomical entity **(Figure 6)**. Gene Set Enrichment Analysis (GSEA) was also done. All gene sets file for *Mus musculus* was downloaded from Broad Institute and GSEA analysis was carried out using fgsea function in R. A table of top 10 upregulated and downregulated pathways was made, and enrichment score plots were plotted for the top three upregulated and downregulated pathways. Interestingly, Hepatic graft versus host disease (GVHD) was downregulated with TMG treatment. LIM_mammary_stem_cell_up and WP_ESC-pluripotency pathways were upregulated **(Figure 6 and Figure 7)**. These pathways are interesting; however, to find more specific pathways related to liver regeneration we need to repeat the experiment with longer duration of TMG injection (one month to six months).

a)

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> go.results
                                                                                Term Ont     N  DE          P.DE
GO:0005622                                                  intracellular anatomical structure  CC 14280 481 4.079381e-52
GO:0005737                                                                           cytoplasm  CC 11464 424 6.406851e-52
GO:0110165                                                             cellular anatomical entity  CC 19113 554 1.721809e-46
GO:0043226                                                                            organelle  CC 12923 439 7.944567e-44
GO:0043229                                                                 intracellular organelle  CC 12611 431 5.112011e-43
GO:0043231                                                 intracellular membrane-bounded organelle  CC 11553 399 2.256641e-38
GO:0043227                                                              membrane-bounded organelle  CC 12021 408 6.536778e-38
GO:0005488                                                                                 binding  MF 14025 420 1.417041e-25
GO:0012505                                                                     endomembrane system  CC  4081 184 1.489375e-25
GO:0009987                                                                            cellular process  BP 17392 481 4.834199e-24
GO:0005783                                                                     endoplasmic reticulum  CC  1827 101 1.900909e-19
GO:0005515                                                                           protein binding  MF  9518 300 7.574739e-18
GO:0065007                                                                      biological regulation  BP 12995 370 4.830556e-16
GO:0031090                                                                        organelle membrane  CC  2275 107 1.270103e-15
GO:0032502                                                                      developmental process  BP  6627 223 3.206694e-15
GO:0098827                                                         endoplasmic reticulum subcompartment  CC   627  48 1.119052e-14
GO:0016020                                                                                membrane  CC  9577 290 1.562703e-14
GO:0005789                                                                  endoplasmic reticulum membrane  CC   614  47 2.149252e-14
GO:0042175                                          nuclear outer membrane-endoplasmic reticulum membrane network  CC   641  48 2.569722e-14
GO:0048856                                                                  anatomical structure development  BP  6150 208 3.022586e-14
GO:0140534                                                       endoplasmic reticulum protein-containing complex  CC   130  22 3.744704e-14
GO:0044237                                                                   cellular metabolic process  BP  9957 297 4.185675e-14
GO:0008152                                                                         metabolic process  BP 11238 323 1.750131e-13
GO:0042221                                                                      response to chemical  BP  4312 158 2.720700e-13
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b)

Pathway	Gene ranks	NES	pval	padj
LIM_MAMMARY_STEM_CELL_UP		1.95	2.3e-09	3.9e-07
WP_ESC_PLURIPOTENCY_PATHWAYS		2.26	2.1e-06	2.4e-04
PLASARI_TGFB1_TARGETS_10HR_UP		2.05	5.5e-06	5.4e-04
H_GRADE_LARGE_CELL_NEUROENDOCRINE_LUNG_CARCINOMA_DN		1.64	2.6e-05	2.1e-03
GUILLAUMOND_KLF10_TARGETS_UP		2.12	4.3e-05	2.6e-03
PASINI_SUZ12_TARGETS_DN		1.64	5.3e-05	3.0e-03
SANDERSON_PPARA_TARGETS		2.02	1.2e-04	5.7e-03
IVANOVA_HEMATOPOIESIS_STEM_CELL		1.81	1.2e-04	5.9e-03
WP_INSULIN_SIGNALING		1.89	1.8e-04	7.9e-03
MCBRYAN_PUBERTAL_TGFB1_TARGETS_UP		1.78	2.1e-04	9.0e-03
YU_MYC_TARGETS_DN		-2.22	7.0e-07	9.5e-05
HEVNER_TELENCEPHALON_MICROGLIA		-2.21	5.9e-07	8.9e-05
QI_PLASMACYTOMA_UP		-2.11	1.7e-09	3.2e-07
GRAESSMANN_RESPONSE_TO_MC_AND_SERUM_DEPRIVATION_UP		-2.21	2.2e-11	4.7e-09
WUNDER_INFLAMMATORY_RESPONSE_AND_CHOLESTEROL_UP		-2.48	1.2e-11	3.0e-09
MARKEY_RB1_ACUTE_LOF_DN		-2.31	1.7e-14	5.0e-12
ZEMEK_IMMUNE_CHECKPOINT_BLOCKADE_OVARIAN_CANCER_RENCA_UP		-2.76	6.6e-20	2.5e-17
ZEMEK_IMMUNE_CHECKPOINT_BLOCKADE_OVARIAN_CANCER_OVERLAP_UP		-2.76	6.6e-20	2.5e-17
ICHIBA_GRAFT_VERSUS_HOST_DISEASE_35D_UP		-2.70	6.2e-21	4.7e-18
ICHIBA_GRAFT_VERSUS_HOST_DISEASE_D7_UP		-2.84	3.6e-26	5.4e-23

Figure 6: Pathway Analysis: a) Significantly regulated pathways as identified by gene ontology (GO) analysis **b) Top 10 upregulated and downregulated pathways** as identified by Gene Set Enrichment Analysis (GSEA).

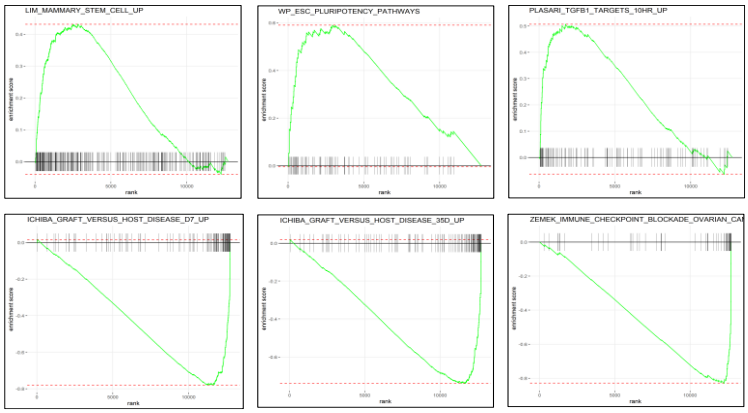

Figure 7: Enrichment Score plot: Enrichment score plots were plotted for the top three upregulated and downregulated pathways in GSEA analysis

Gene Clustering Analysis: Hierarchical clustering and k-means clustering analysis was done to explore if the genes display any clustering pattern and the genes in clusters were assessed using CoxPH analysis. Different k values were used. Here k=3 for hierarchical clustering and k=3 and k=2 for k-means clustering is shown. However, there was no specific clustering of genes present with either of these methods. Hence, a heatmap of genes was plotted to see if the genes display any common expression pattern between the control and treatment groups. There was no significant clustering pattern seen with the heatmap as well **(Figure 8)**. These data indicate that RNA-Seq analysis needs to be repeated with longer periods of injection (1 month and 6-month injections) along with mice having liver specific OGT KO and OGA KO for comparison. Additionally, subject 3 had lower gene expression compared to the remaining subjects. The problem with subject 3 needs to be analyzed further. At least 5 mice of control and 5 mice of treatment will be used for further analysis.

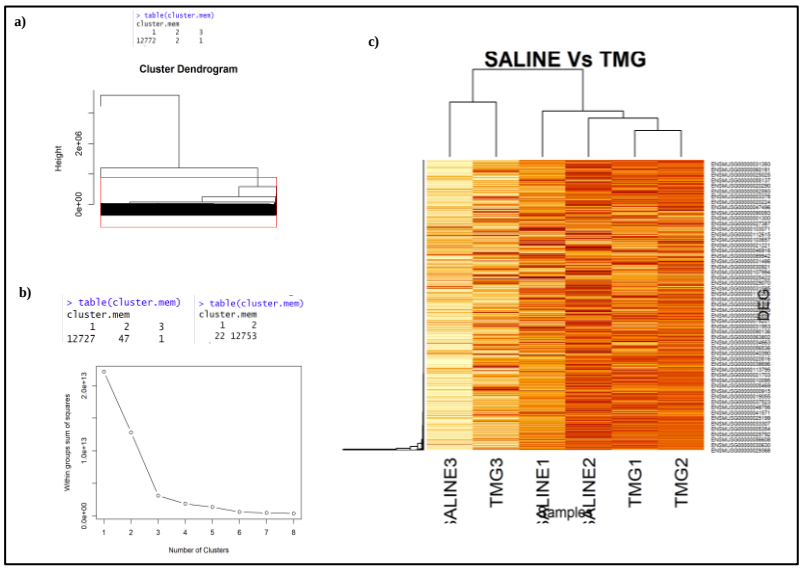

Figure 8: Gene Clustering Analysis: a) Hierarchical clustering dendogram with k=3 **b)** k-means clustering with k=3 and k=2 **c)** Heatmap to see if the genes exhibit clustering patterns between the Saline and TMG mice groups

Discussion and Conclusion: O-GlcNAcylation is a dynamic post-translational modification where a single N-acetylglucosamine (GlcNAc) sugar dynamically cycles on and off serine and threonine residues of proteins by the action of the enzymes OGT and OGA respectively (3). Disrupted O-GlcNAc homeostasis affects various functions including cell proliferation, differentiation, and disease progression (4). Here we study how disrupted O-GlcNAcylation affects liver regeneration. Decreased O-GlcNAcylation was found to disrupt liver regeneration by promoting hepatocyte proliferating, eventually forming liver tumors. This phenotype was reversed in liver-specifc OGA KO mice (7). Thus, in this study we increased O-GlcNAcylation via Thiamet-G (TMG - OGA inhibitor) treatment. If OGA is inhibited, O-GlcNAC levels increase which could promote liver regeneration. We wanted to further investigate this mechanism and identify the genes that could be involved in this process.

We did a pilot study of RNA Sequencing Analysis with mice injected for two weeks with TMG. Three mice were injected intraperitoneally for two weeks with TMG and three mice were injected with saline for control. Thus we had two groups of mice (Saline and TMG) with three subjects each (10). Quality control plots revealed subject 3 of both groups have a grouping effect and decreased library size. So, normalization was carried out using TMM method after filtering out genes with gene counts as zero for all samples and keeping only genes with counts per million (cpm) greater than or equal to 2 to reduce the burden of multiple testing in differential gene expression analyses (13). TMM normalization method is based on the hypothesis that most genes are not differentially expressed (DE). In edgeR, the TMM normalization method computes normalization factors that represent sample-specific biases. These factors are multiplied by library size to yield the effective library size, i.e., the library size that would be obtained if those biases were not present. The effective library sizes can then be used for various normalization purposes, commonly as offsets in generalized linear models (14). However, even after normalization some grouping was found with TMG sample 3 and the library size of sample 3 saline and sample 3 TMG looked smaller than the rest **(Figure 4)**.

Differential gene expression analysis was carried out for the filtered data using EdgeR via Exact test as there are only two groups – Saline and TMG injection. For data with more than two groups, glmfit can be used instead of EdgeR (4, 7, 11). However, out of 12775 genes analyzed, only 19 genes were downregulated, 10 genes were upregulated, and 12,746 genes had no significant changes in expression. Some of the top significantly regulated genes included Plin4, SELENBP2, MANF and OGA. Plin4 plays a role in triacylglycerol packaging into adipocytes (15). SELENBP2 gene encodes a member of the selenium-binding protein family. Selenium is an essential nutrient that exhibits potent anticarcinogenic properties. Decreased expression of this gene may be associated with several types of cancer (16). Reducing expression of MANF increases susceptibility to ER stress-induced death and results in cell proliferation (17). TMG is an OGA inhibitor, however OGA is upregulated with

TMG treatment because too much increase in O-GlcNAc is detrimental to the cells. So, the cells by alternative mechanisms increase OGA to prevent further increase in O-GlcNAc (3).

These genes are interesting however to find if any genes are related specifically to liver regeneration, we did pathway analysis (4, 11). GO pathway analysis showed some significantly enriched pathways including intracellular anatomical structure, cytoplasm and cellular anatomical entity (18). Interestingly, GSEA analysis showed Hepatic graft versus host disease (GVHD) was downregulated with TMG treatment. LIM_mammary_stem_cell_up and WP_ESC-pluripotency pathways were upregulated (19). These pathways are interesting; however, we did not find specific pathways related to liver regeneration. Hence, the analysis must be repeated with longer duration of TMG injection (at least one month or up to six months TMG injection (9)) and with liver-specific genetic OGT knockout (KO) and OGA KO mice (7).

Furthermore, hierarchical clustering and k-means clustering analysis to explore if the genes display any clustering pattern revealed no specific clustering gene patterns to be present (11). Hierarchical clustering groups similar objects into groups called clusters. The endpoint is a set of clusters, where each cluster is distinct from other clusters, and the objects within each cluster are broadly similar to each other (20). Hierarchical clustering is the most over-used statistical method in gene expression analysis. It tends to be pretty unstable (11). K-means clustering aims to partition the observations into clusters in which each observation belongs to the cluster with the nearest mean, serving as a prototype of the cluster (11, 21). K-means clustering with k=2 showed 22 genes in one group and the remaining in the other group, which could indicate the differentially expressed genes are grouping together. However, heatmap showed that there was no significant clustering of genes between the saline and TMG groups. Additionally, subject 3 had lower gene expression compared to the remaining subjects. The problem with subject 3 needs to be analyzed further.

Together, this RNA-Seq data analysis indicates that the RNA-Seq analysis must be repeated with longer periods of injection (1 month and 6 month injections) along with mice having liver specific OGT KO and OGA KO for effective comparison (7, 9). At least 5 mice of control and 5 mice of treatment group will be used for further analysis to increase statistical power. Repeating RNA-Seq analysis with these changes will lead to the identification of genes with significant differential expression due to O-GlcNAc disruptions in mice liver. This in turn will help to identify the genes involved in the regulation of liver regeneration by O-GlcNAc.

References:

1. Zachara NE, Hart GW. O-GlcNAc a sensor of cellular state: the role of nucleocytoplasmic glycosylation in modulating cellular function in response to nutrition and stress. Biochimica et Biophysica Acta (BBA)-General Subjects. 2004;1673(1-2):13-28.
2. Hart GW, Slawson C, Ramirez-Correa G, Lagerlof O. Cross talk between O-GlcNAcylation and phosphorylation: roles in signaling, transcription, and chronic disease. Annual review of biochemistry. 2011;80:825-58.
3. Zhang Z, Tan EP, VandenHull NJ, Peterson KR, Slawson C. O-GlcNAcase expression is sensitive to changes in O-GlcNAc homeostasis. Frontiers in endocrinology. 2014;5:206.
4. Tan EP, McGreal SR, Graw S, Tessman R, Koppel SJ, Dhakal P, et al. Sustained O-GlcNAcylation reprograms mitochondrial function to regulate energy metabolism. Journal of Biological Chemistry. 2017;292(36):14940-62.
5. Machacek M, Zhang Z, Tan EP, Li J, Li T, Villar MT, et al. Elevated O-GlcNAcylation Enhances Pro-Inflammatory Th17 Function by Altering the Lipid Microenvironment. bioRxiv. 2018:305722.
6. Wang Z, Udeshi ND, Slawson C, Compton PD, Sakabe K, Cheung WD, et al. Extensive crosstalk between O-GlcNAcylation and phosphorylation regulates cytokinesis. Science signaling. 2010;3(104):ra2-ra.
7. Robarts DR, McGreal SR, Umbaugh DS, Parkes WS, Kotulkar M, Abernathy S, et al. Regulation of liver regeneration by hepatocyte O-GlcNAcylation in mice. Cellular and Molecular Gastroenterology and Hepatology. 2022;13(5):1510-29.
8. Yuzwa SA, Shan X, Macauley MS, Clark T, Skorobogatko Y, Vosseller K, et al. Increasing O-GlcNAc slows neurodegeneration and stabilizes tau against aggregation. Nature chemical biology. 2012;8(4):393-9.
9. Ephrame SJ, Cork GK, Marshall V, Johnston MA, Shawa J, Alghusen I, et al. O-GlcNAcylation regulates extracellular signal-regulated kinase (ERK) activation in Alzheimer's disease. Frontiers in Aging Neuroscience. 2023;15.
10. Macauley MS, Vocadlo DJ. Increasing O-GlcNAc levels: An overview of small-molecule inhibitors of O-GlcNAcase. Biochimica et Biophysica Acta (BBA)-General Subjects. 2010;1800(2):107-21.
11. Chalise Dp. BIOS-799-STATISTICAL GENOMICS. 2023.
12. heatmap [Available from: https://r-graph-gallery.com/heatmap

13. Dillies M-A, Rau A, Aubert J, Hennequet-Antier C, Jeanmougin M, Servant N, et al. A comprehensive evaluation of normalization methods for Illumina high-throughput RNA sequencing data analysis. Briefings in bioinformatics. 2013;14(6):671-83.
14. TMM Normalization [Available from: https://support.bioconductor.org/p/79379/#:~:text=In%20edgeR%2C%20the%20TMM%20method,those%20biases%20were%20not%20present

15. PLIN4 [Available from: https://www.genecards.org/cgi-bin/carddisp.pl?gene=PLIN4.
16. SELENBP2 [Available from: https://www.genecards.org/cgi-bin/carddisp.pl?gene=SELENBP1&keywords=Selenbp2.
17. MANF [Available from: https://www.genecards.org/cgi-bin/carddisp.pl?gene=MANF#:~:text=GeneCards%20Summary%20for%20MANF%20Gene,to%20elevated%20platelet%20cytosolic%20Ca2%2B.
18. GO Pathway Analysis [Available from: http://geneontology.org/.
19. GSEA Pathway Analysis [Available from: https://www.gsea-msigdb.org/gsea/index.jsp.
20. Hierarchical clustering [Available from: https://www.displayr.com/what-is-hierarchical-clustering/#:~:text=Hierarchical%20clustering%2C%20also%20known%20as,broadly%20similar%20to%20each%20other.
21. K-Means Clustering [Available from: https://en.wikipedia.org/wiki/K-means_clustering.

R Code:

```
###-------------------------------------------------------------------------------------------
###-------------------------------------------------------------------------------------------
###-------------------------------------------------------------------------------------------


#-------------------------------------------------------------------------------
#TEST USING EXACT METHOD
#-------------------------------------------------------------------------------


#Gene Set Enrichment analysis using fgsea package in R
# https://bioconductor.org/packages/release/bioc/html/fgsea.html
# http://bioconductor.org/packages/devel/bioc/vignettes/fgsea/inst/doc/fgsea-tutorial.html
# https://github.com/ctlab/fgsea

####-----------------------------------------------
#### fgsea Example
####-----------------------------------------------
rm(list=ls())
 if (!requireNamespace("BiocManager", quietly = TRUE))
 install.packages("BiocManager")
 BiocManager::install("fgsea")
library(data.table)
library(fgsea)
library(ggplot2)
library(edgeR)
library(RColorBrewer)


### Load Data example
load("C:\\Users\\Sophiya John\\Desktop\\J_STATISTICS\\J_STATISTICAL GENOMICS BIOS 799\\GSEAexample.RData")
class(dat)
head(dat)
colnames(dat)
dim(dat)
# [1] 20382    22
dat <- dat[1:round(nrow(dat)/4),]   ## Select 1/4th of the genes for demonstration
dim(dat)                            ## Using all data run time will be longer
```

```
dat

#mydata

rnaseq <- read.csv("C:\\Users\\Sophiya John\\Desktop\\J_STATISTICS\\J_STATISTICAL GENOMICS BIOS 799\\RNASeq2.txt", header=T, sep="\t")
#rnaseq
names(rnaseq)
#Choose only saline TMG
df <- rnaseq[ -c(3:11) ]
df2 <- df[ -c(8:10) ]
names(df2)
df2<-as.data.frame(df2)
df2
#Change col names
colnames(df2) <- c('GENES','TMG1','TMG2','TMG3','SALINE1','SALINE2','SALINE3')
#RELOCATE SALINE TO FIRST 3 COLUMNS
df2
library(dplyr)
TMG2 <- df2 %>%
  relocate(SALINE1, .before = TMG1)%>%
  relocate(SALINE2, .before = TMG1)%>%
  relocate(SALINE3, .before = TMG1)

TMG2

#TO SEPARATE ENSEMBL ID AND GENE NAME

library(dplyr)
library(tidyr)

TMG1<-TMG2 %>% separate(GENES, c('ENSEMBLID', 'GENENAME'))
TMG1
TMG<-as.data.frame(TMG1)
TMG<-TMG[,-2]
# Assign the 'ID' column as row names
rownames(TMG) <- TMG$ENSEMBLID

# Remove the 'ID' column from the data frame (if needed)
TMG$ENSEMBLID <- NULL
```

```
TMG
dim(TMG)
#[1] 55298     6

dat
dim(dat)
#[1] 5096   22

###---------------------------
### DE analysis using edgeR
###---------------------------
### Genes filltering based on counts
dat <- dat[rowSums(dat)!=0,]     ## Remove if gene counts is zero for all samples
dim(dat)
#[1] 5096   22

TMG <- TMG[rowSums(TMG)!=0,]     ## Remove if gene counts is zero for all samples
dim(TMG)
#[1] 25705     6

### MDS (Multidimensional Scaling) plot:
plotMDS(dat, main="Multipledimensional scaling plot (MDS)")
plotMDS(TMG, main="Multipledimensional scaling plot (MDS)")


### Mean Variance plot by gene
###----------------------------------------------
mean.x <- apply(dat,1,mean)
var.x <- apply(dat,1,var)
plot(log10(mean.x),log10(var.x),pch=20, xlab="Mean (in log 10 scale)",ylab="Variance (in log 10 scale)",
    xlim=c(0,9),ylim=c(0,9),main="Variance vs Mean")
abline(0,1,col=3,lwd=2)


mean.x <- apply(TMG,1,mean)
var.x <- apply(TMG,1,var)
plot(log10(mean.x),log10(var.x),pch=20, xlab="Mean (in log 10 scale)",ylab="Variance (in log 10 scale)",
    xlim=c(0,9),ylim=c(0,9),main="Variance vs Mean for TMG data")
abline(0,1,col=3,lwd=2)
```

```
### Barplot
###-----------------------------------------------
lib.size <- colSums(dat)
barplot(lib.size,xaxt="n",xlab="Study samples",ylab="Library size values",main="Barplot of library size",
	col=c(rep("lightgreen",11),rep("lightcoral",11)),sub="(Red line represents mean)")
abline(h=mean(lib.size),lwd=2,col="red")
legend("topright",c("Normal","Tumor"),lty=c(1,1),lwd=c(4,4),col=c("lightgreen","red"),bty="o",cex=1,text.font=2)


lib.size <- colSums(TMG)
barplot(lib.size,xaxt="n",xlab="Study samples",ylab="Library size values",main="Barplot of library size",
	col=c(rep("lightgreen",3),rep("lightcoral",3)),sub="(Red line represents mean)")
abline(h=mean(lib.size),lwd=2,col="red")
legend("topright",c("SALINE","TMG"),lty=c(1,1),lwd=c(4,4),col=c("lightgreen","red"),bty="o",cex=1,text.font=2)


### Boxplot (In log10 scale)
###-----------------------------------------------
boxplot(x=as.list(as.data.frame(log10(dat+1))),xlab="Samples",ylab="Values in log10 scale",main="Boxplot across study samples",
	col=c(rep("lightgreen",11),rep("lightcoral",11)))
legend("topright",c("Normal","Tumor"),lty=c(1,1),lwd=c(4,4),col=c("lightgreen","red"),bty="o",cex=1,text.font=2)


boxplot(x=as.list(as.data.frame(log10(TMG+1))),xlab="Samples",ylab="Values in log10 scale",main="Boxplot across study samples",
	col=c(rep("lightgreen",3),rep("lightcoral",3)))
legend("topright",c("SALINE","TMG"),lty=c(1,1),lwd=c(4,4),col=c("lightgreen","red"),bty="o",cex=1,text.font=2)


### Density plots by sample
###-----------------------------------------------
```

```
for (i in 1:ncol(dat)){
  if (i==1){
    plot(density(log10(dat[,1])),main="Density plot across study samples",xlab="Subjects",col="gray",ylim=c(0,0.8))}
  else {
    den <- density(log10(dat[,i]))
    lines(den$x,den$y,col="gray")}
}



for (i in 1:ncol(TMG)){
  if (i==1){
    plot(density(log10(TMG[,1])),main="Density plot across study samples for TMG data",xlab="Subjects",col="gray",ylim=c(0,0.8))}
  else {
    den <- density(log10(TMG[,i]))
    lines(den$x,den$y,col="gray")}
}


## edgeR DE analysis
##-----------------------------------------------------------------------------------------
keep <- rowSums(cpm(as.matrix(dat))>1)>=2
# At least 2 samples have to have cpm > 1.
data.filtered <- dat[keep,]
dim(data.filtered)
rm(keep)
d <- DGEList(counts=as.matrix(data.filtered), lib.size=colSums(data.filtered), group=c(rep("Normal",11),rep("Tumor",11)))
dim(d)
d <- calcNormFactors(d, method="TMM")  ## Calculates normalization factors
d <- estimateDisp(d)  ## Calculates genewise dispersion parameter adjusted using bayesian empirical method(Recommended, edgeR)
d

keep1 <- rowSums(cpm(as.matrix(TMG))>1)>=2
# At least 2 samples have to have cpm > 1.
data.filtered1 <- TMG[keep1,]
```

```
dim(data.filtered1)
#[1] 12775     6
rm(keep1)
d1 <- DGEList(counts=as.matrix(data.filtered1), lib.size=colSums(data.filtered1),
group=c(rep("SALINE",3),rep("TMG",3)))
dim(d1)
d1
d1 <- calcNormFactors(d1, method="TMM")       ## Calculates normalization factors
d1
d1 <- estimateDisp(d1)                        ## Calculates genewise dispersion parameter adjusted using
bayesian empirical method(Recommended, edgeR)
d1



## BCV
plotBCV(d)
plotBCV(d1)

de.test <- exactTest(d,pair=c("Normal","Tumor")) ## First entry under pair is the baseline
de.test.FDR <- topTags(de.test,n=Inf,adjust.method="BH", sort.by="PValue")
head(de.test.FDR$table)
summary(de <- decideTestsDGE(de.test, p=0.05, adjust="BH"))  ## Counts up and down regulated genes

de.test1 <- exactTest(d1,pair=c("SALINE","TMG")) ## First entry under pair is the baseline
de.test1.FDR <- topTags(de.test1,n=Inf,adjust.method="BH", sort.by="PValue")
head(de.test1.FDR$table)
summary(de1 <- decideTestsDGE(de.test1, p=0.05, adjust="BH"))  ## Counts up and down regulated
genes
dim(d1)

### MDS (Multidimensional Scaling) plot:
plotMDS(d1, main="Multipledimensional scaling plot (MDS)")



### Mean Variance plot by gene
###----------------------------------------------

mean.x <- apply(d1,1,mean)
var.x <- apply(d1,1,var)
```

```
plot(log10(mean.x),log10(var.x),pch=20, xlab="Mean (in log 10 scale)",ylab="Variance (in log 10 scale)",
    xlim=c(0,9),ylim=c(0,9),main="Variance vs Mean for TMG data")
abline(0,1,col=3,lwd=2)



### Barplot
###-----------------------------------------------
d2<-as.matrix(d1)
lib.size <- colSums(d2)
d2
d1
barplot(lib.size,xaxt="n",xlab="Study samples",ylab="Library size values",main="Barplot of library size",
      col=c(rep("lightgreen",3),rep("lightcoral",3)),sub="(Red line represents mean)")
abline(h=mean(lib.size),lwd=2,col="red")
legend("topright",c("SALINE","TMG"),lty=c(1,1),lwd=c(4,4),col=c("lightgreen","red"),bty="o",cex=1,text.font=2)


### Boxplot (In log10 scale)
###-----------------------------------------------
boxplot(x=as.list(as.data.frame(log10(dat+1))),xlab="Samples",ylab="Values in log10 scale",main="Boxplot across study samples",
      col=c(rep("lightgreen",11),rep("lightcoral",11)))
legend("topright",c("Normal","Tumor"),lty=c(1,1),lwd=c(4,4),col=c("lightgreen","red"),bty="o",cex=1,text.font=2)


boxplot(x=as.list(as.data.frame(log10(d2+1))),xlab="Samples",ylab="Values in log10 scale",main="Boxplot across study samples",
      col=c(rep("lightgreen",3),rep("lightcoral",3)))
legend("topright",c("SALINE","TMG"),lty=c(1,1),lwd=c(4,4),col=c("lightgreen","red"),bty="o",cex=1,text.font=2)


### Density plots by sample
###-----------------------------------------------

for (i in 1:ncol(dat)){
```

```
  if (i==1){
    plot(density(log10(dat[,1])),main="Density plot across study samples",xlab="Subjects",col="gray",ylim=c(0,0.8))}
  else {
    den <- density(log10(dat[,i]))
    lines(den$x,den$y,col="gray")}
}



for (i in 1:ncol(d2)){
  if (i==1){
    plot(density(log10(d2[,1])),main="Density plot across study samples for TMG data",xlab="Subjects",col="gray",ylim=c(0,0.8))}
  else {
    den <- density(log10(d2[,i]))
    lines(den$x,den$y,col="gray")}
}


# ### plotSmear
# ###-------------------------------------------------------
detags <- rownames(d)[as.logical(de)]
plotSmear(de.test, de.tags=detags,main="Smear Plot",sub="(The horizontal blue lines show 4-fold changes)",cex=1)
abline(h = c(-2, 2), col = "blue")

detags1 <- rownames(d1)[as.logical(de1)]
plotSmear(de.test1, de.tags=detags1,main="Smear Plot",sub="(The horizontal blue lines show 4-fold changes)",cex=1)
abline(h = c(-2, 2), col = "blue")


### Volcano plot
###-------------------------------------------------------
d.volcano <-  de.test.FDR$table[,c("logFC","PValue")]
par(mar=c(5,4,4,5))
plot(d.volcano$logFC, -log(d.volcano$PValue,10), main="",pch=20, cex=2,xlab=expression(log[2]~fold~change), ylab=expression(-log[10]~pvalue),
    xlim=c(min(d.volcano$logFC)-1,max(d.volcano$logFC)+1))
```

```
title("Volcano plot")


# Log2 fold change and p-value cutoff
lfc <- 2
pval <- 0.05
# Selecting interesting genes
sigGenes <- (abs(d.volcano$logFC)> lfc & -log(d.volcano$PValue,10) > -log10(pval))
# Identifying the selected genes
points(d.volcano[sigGenes,]$logFC,-log(d.volcano[sigGenes,]$PValue,10),pch=20,col="red",cex=2)
abline(h=-log10(pval),col="green3",lty=2)
abline(v=c(-lfc,lfc),col="blue",lty=2)
mtext(paste("pval = ",round(pval,2)),side=4,at=-log10(pval),cex=0.8,line=0.5,las=1)
mtext(c(paste("-",lfc,"fold"),paste("+",lfc,"fold")),side=3,at=c(-lfc,lfc),cex=1,line=0.2)


d.volcano1 <-  de.test1.FDR$table[,c("logFC","PValue")]
par(mar=c(5,4,4,5))
plot(d.volcano1$logFC,                    -log(d.volcano1$PValue,10),                    main="",pch=20,
cex=2,xlab=expression(log[2]~fold~change), ylab=expression(-log[10]~pvalue),
    xlim=c(min(d.volcano1$logFC)-1,max(d.volcano1$logFC)+1))
title("Volcano plot for TMG data")

# Log2 fold change and p-value cutoff
lfc1 <- 2
pval1 <- 0.05
# Selecting interesting genes
sigGenes1 <- (abs(d.volcano1$logFC)> lfc1 & -log(d.volcano1$PValue,10) > -log10(pval1))
# Identifying the selected genes
points(d.volcano1[sigGenes1,]$logFC,-log(d.volcano1[sigGenes1,]$PValue,10),pch=20,col="red",cex=2)
abline(h=-log10(pval1),col="green3",lty=2)
abline(v=c(-lfc1,lfc1),col="blue",lty=2)
mtext(paste("pval = ",round(pval1,2)),side=4,at=-log10(pval1),cex=0.8,line=0.5,las=1)
mtext(c(paste("-",lfc1,"fold"),paste("+",lfc1,"fold")),side=3,at=c(-lfc1,lfc1),cex=1,line=0.2)


d.edgeR.result <- de.test.FDR$table
head(d.edgeR.result)

d1.edgeR.result <- de.test1.FDR$table
```

```
head(d1.edgeR.result)



# Map Ensembl gene IDs to symbol. First create a mapping table.
library(biomaRt)
mart <- useDataset("hsapiens_gene_ensembl", useMart("ensembl"))
ensg.id <- rownames(d.edgeR.result)
d.gene.id <- getBM(filters= "ensembl_gene_id", attributes= c("ensembl_gene_id","hgnc_symbol"),
values=ensg.id, mart= mart)
dim(d.gene.id)
head(d.gene.id)
d.gene.id <- subset(d.gene.id,hgnc_symbol!="")  # Remove the ENSG ids that do not have hgnc_symbol
rownames(d.gene.id) <- d.gene.id$ensembl_gene_id
dim(d.gene.id)



library(biomaRt)
mart <- useDataset("mmusculus_gene_ensembl", useMart("ensembl"))
ensg.id <- rownames(d1.edgeR.result)
d.gene.id <- getBM(filters= "ensembl_gene_id", attributes= c("ensembl_gene_id","mgi_symbol"),
values=ensg.id, mart=mart)
#attributes=mgi_symbol for mice
dim(d.gene.id)
head(d.gene.id)
d.gene.id <- subset(d.gene.id,mgi_symbol!="")  # Remove the ENSG ids that do not have mgi_symbol
rownames(d.gene.id) <- d.gene.id$ensembl_gene_id
dim(d.gene.id)
head(d.gene.id)

listEnsembl()
ensembl <- useEnsembl(biomart = "genes")
ensembl
datasets <- listDatasets(ensembl)
head(datasets)
searchDatasets(mart = ensembl, pattern = "mmusculus")


###Combine the datasets together
```

```
common.genes <- intersect(rownames(d.edgeR.result),rownames(d.gene.id))
length(common.genes)
length(unique(common.genes))

common.genes <- intersect(rownames(d1.edgeR.result),rownames(d.gene.id))
length(common.genes)
length(unique(common.genes))

# Merge gene id data with the results data
d.tmp1 <- d.gene.id[common.genes,]
d.tmp2 <- d.edgeR.result[common.genes,]
all(rownames(d.tmp1)==rownames(d.tmp2))

d.tmp1 <- d.gene.id[common.genes,]
d.tmp2 <- d1.edgeR.result[common.genes,]
all(rownames(d.tmp1)==rownames(d.tmp2))

# TRUE
d.tmp <- cbind(d.tmp1,d.tmp2)
head(d.tmp)
d.tmp <- d.tmp[order(d.tmp$PValue),] ## Sort the data in the order of significance
head(d.tmp)

d.tmp <- cbind(d.tmp1,d.tmp2)
head(d.tmp)
d.tmp <- d.tmp[order(d.tmp$PValue),] ## Sort the data in the order of significance
head(d.tmp)



# Prepare the gene list
gene.list <- d.tmp$logFC            # rank-ordered gene list
names(gene.list) <- d.tmp$hgnc_symbol

gene.list <- d.tmp$logFC            # rank-ordered gene list
names(gene.list) <- d.tmp$mgi_symbol

# Barplot of ranked fold changes
barplot(sort(gene.list, decreasing = T),axisnames=FALSE,main="Plot of ranked gene Fold changes")

barplot(sort(gene.list, decreasing = T),axisnames=FALSE,main="Plot of ranked gene Fold changes")
```

```
###GO analysis
library(org.Hs.eg.db)
library(edgeR)
library(GO.db)
d.go <- d.tmp
#d.go.DE <- subset(d.go,FDR<0.05)
d.go.DE <- subset(d.go,PValue<0.05)
d.entrez.id <- mapIds(org.Hs.eg.db,
keys=d.go.DE$hgnc_symbol,column="ENTREZID",keytype="SYMBOL")
length(d.entrez.id)
head(d.entrez.id)
all(d.go.DE$hgnc_symbol==names(d.entrez.id))
go.test <- goana(d.entrez.id,species="Hs")
go.results <- topGO(go.test, sort = "DE", number = Inf)
head(go.results)
sum(go.results$P.DE<10^(-5))


library(org.Mm.eg.db)
library(edgeR)

if (!require("BiocManager", quietly = TRUE))
  install.packages("BiocManager")

BiocManager::install("GO.db")

library(GO.db)
d.go <- d.tmp
#d.go.DE <- subset(d.go,FDR<0.05)
d.go.DE <- subset(d.go,PValue<0.05)
#attributes=mgi_symbol for mice
d.entrez.id <- mapIds(org.Mm.eg.db,
keys=d.go.DE$mgi_symbol,column="ENTREZID",keytype="SYMBOL")
length(d.entrez.id)
head(d.entrez.id)
all(d.go.DE$mgi_symbol==names(d.entrez.id))
go.test <- goana(d.entrez.id,species="Mm")
go.results <- topGO(go.test, sort = "DE", number = Inf)
head(go.results)
```

```
sum(go.results$P.DE<10^(-5))

###KEGG analysis
d.kegg <- d.tmp
#d.kegg.DE <- subset(d.kegg,FDR<0.05)
d.kegg.DE <- subset(d.kegg,PValue<0.05)
all(d.kegg.DE$hgnc_symbol==names(d.entrez.id))

kegg.test <- kegga(d.entrez.id,species="Hs")
kegg.results <- topKEGG(kegg.test, sort = "DE", number = Inf)
head(kegg.results)
sum(kegg.results$P.DE<10^(-5))



d.kegg <- d.tmp
#d.kegg.DE <- subset(d.kegg,FDR<0.05)
d.kegg.DE <- subset(d.kegg,PValue<0.05)
all(d.kegg.DE$mgi_symbol==names(d.entrez.id))
d.entrez.id <- mapIds(org.Mm.eg.db,
keys=d.go.DE$mgi_symbol,column="ENTREZID",keytype="SYMBOL")
length(d.entrez.id)
head(d.entrez.id)



if (!require("BiocManager", quietly = TRUE))
  install.packages("BiocManager")

BiocManager::install("KEGGREST")
library(KEGGREST)
kegg.test <- kegga(d.entrez.id,species="Mm")
kegg.results <- topKEGG(kegg.test, sort = "DE", number = Inf)
head(kegg.results)
sum(kegg.results$P.DE<10^(-5))



### GSEA analysis
# Load All gene sets file downloaded from Broad Institute
# The following website contains the gene set collection or the complete Molecular Signatures Database
(MSigDB)
```

```
# http://software.broadinstitute.org/gsea/downloads.jsp#msigdb
#
all.gene.sets <- gmtPathways("C:\\Users\\Sophiya John\\Desktop\\J_STATISTICS\\J_STATISTICAL GENOMICS BIOS 799\\msigdb.v7.4.symbols.gmt")
class(all.gene.sets)
length(all.gene.sets)
all.gene.sets[1:2]
# Show first a few pathways, and within those, show only the first few genes.
library(tidyverse)
all.gene.sets %>% head() %>% lapply(head)

### Now run fgsea
fgseaRes <- fgsea(pathways = all.gene.sets,stats = gene.list,minSize=15,maxSize=500,eps=0)
head(fgseaRes)
head(fgseaRes[order(pval), ])
sum(fgseaRes[, padj < 0.05])
# Make a few Enrichment Plots
#Plot1
plotEnrichment(all.gene.sets[["GAO_LARGE_INTESTINE_ADULT_CI_MESENCHYMAL_CELLS"]]
,gene.list) + labs(title="GAO_LARGE_INTESTINE_ADULT_CI_MESENCHYMAL_CELLS")
#Plot2
plotEnrichment(all.gene.sets[["CHEN_METABOLIC_SYNDROM_NETWORK"]],gene.list)  +
labs(title="CHEN_METABOLIC_SYNDROM_NETWORK")
tail(fgseaRes[order(pval), ])
#Plot3
plotEnrichment(all.gene.sets[["ZNF610_TARGET_GENES"]],gene.list)  +
labs(title="ZNF610_TARGET_GENES")
#Plot4
plotEnrichment(all.gene.sets[["ZSCAN5DP_TARGET_GENES"]],gene.list)  +
labs(title="ZSCAN5DP_TARGET_GENES")
fgseaRes[order(pval), ][15:20,]
#Plot5
plotEnrichment(all.gene.sets[["GOBP_RESPONSE_TO_BIOTIC_STIMULUS"]],gene.list)  +
labs(title="GOBP_RESPONSE_TO_BIOTIC_STIMULUS")

# Make a table plot for a bunch of selected pathways:
topPathwaysUp <- fgseaRes[ES > 0][head(order(pval), n=10), pathway]
topPathwaysDown <- fgseaRes[ES < 0][head(order(pval), n=10), pathway]
topPathways <- c(topPathwaysUp, rev(topPathwaysDown))
plotGseaTable(all.gene.sets[topPathways], gene.list, fgseaRes,gseaParam = 0.5)
```

```
### GSEA analysis
# Load All gene sets file downloaded from Broad Institute
# The following website contains the gene set collection or the complete Molecular Signatures Database (MSigDB)
# http://software.broadinstitute.org/gsea/downloads.jsp#msigdb
#
#mydata

all.gene.sets <- gmtPathways("C:\\Users\\Sophiya John\\Desktop\\J_STATISTICS\\J_STATISTICAL GENOMICS BIOS 799\\m2.all.v2023.1.Mm.symbols.gmt")	#m2curated gene sets
class(all.gene.sets)
length(all.gene.sets)
all.gene.sets[1:2]
# Show first a few pathways, and within those, show only the first few genes.
library(tidyverse)
all.gene.sets %>% head() %>% lapply(head)

### Now run fgsea
fgseaRes <- fgsea(pathways = all.gene.sets,stats = gene.list,minSize=15,maxSize=500,eps=0)
head(fgseaRes)
head(fgseaRes[order(pval), ])
sum(fgseaRes[, padj < 0.05])
# Make a few Enrichment Plots
#Plot1
plotEnrichment(all.gene.sets[["ICHIBA_GRAFT_VERSUS_HOST_DISEASE_D7_UP"]],gene.list) + labs(title="ICHIBA_GRAFT_VERSUS_HOST_DISEASE_D7_UP")
#Plot2
plotEnrichment(all.gene.sets[["ICHIBA_GRAFT_VERSUS_HOST_DISEASE_35D_UP"]],gene.list) + labs(title="ICHIBA_GRAFT_VERSUS_HOST_DISEASE_35D_UP")
#Plot3
plotEnrichment(all.gene.sets[["ZEMEK_IMMUNE_CHECKPOINT_BLOCKADE_OVARIAN_CANCER_OVERLAP_UP"]],gene.list) + labs(title="ZEMEK_IMMUNE_CHECKPOINT_BLOCKADE_OVARIAN_CANCER_OVERLAP_UP")
#Plot4
plotEnrichment(all.gene.sets[["MARKEY_RB1_ACUTE_LOF_DN"]],gene.list) + labs(title="MARKEY_RB1_ACUTE_LOF_DN")
```

```
tail(fgseaRes[order(pval), ])
#Plot5
plotEnrichment(all.gene.sets[["REACTOME_VESICLE_MEDIATED_TRANSPORT"]],gene.list)        +
labs(title="REACTOME_VESICLE_MEDIATED_TRANSPORT")
#Plot6
plotEnrichment(all.gene.sets[["REACTOME_VLDLR_INTERNALISATION_AND_DEGRADATION"]
],gene.list) + labs(title="REACTOME_VLDLR_INTERNALISATION_AND_DEGRADATION")
#Plot7
plotEnrichment(all.gene.sets[["SANSOM_APC_MYC_TARGETS"]],gene.list)                      +
labs(title="SANSOM_APC_MYC_TARGETS")

fgseaRes[order(pval), ][15:20,]
#Plot8
plotEnrichment(all.gene.sets[["PLASARI_TGFB1_TARGETS_10HR_UP"]],gene.list)               +
labs(title="PLASARI_TGFB1_TARGETS_10HR_UP")
#Plot9
plotEnrichment(all.gene.sets[["WP_ELECTRON_TRANSPORT_CHAIN"]],gene.list)                 +
labs(title="WP_ELECTRON_TRANSPORT_CHAIN")
#Plot10
plotEnrichment(all.gene.sets[["MORI_LARGE_PRE_BII_LYMPHOCYTE_DN"]],gene.list)            +
labs(title="MORI_LARGE_PRE_BII_LYMPHOCYTE_DN")

# Make a table plot for a bunch of selected pathways:
topPathwaysUp <- fgseaRes[ES > 0][head(order(pval), n=10), pathway]
topPathwaysDown <- fgseaRes[ES < 0][head(order(pval), n=10), pathway]
topPathways <- c(topPathwaysUp, rev(topPathwaysDown))
plotGseaTable(all.gene.sets[topPathways], gene.list, fgseaRes,gseaParam = 0.5)

#Top pathways UP
#Plot11
plotEnrichment(all.gene.sets[["LIM_MAMMARY_STEM_CELL_UP"]],gene.list)                    +
labs(title="LIM_MAMMARY_STEM_CELL_UP")
#Plot12
plotEnrichment(all.gene.sets[["WP_ESC_PLURIPOTENCY_PATHWAYS"]],gene.list)                +
labs(title="WP_ESC_PLURIPOTENCY_PATHWAYS")
#Plot13
```

```
plotEnrichment(all.gene.sets[["PLASARI_TGFB1_TARGETS_10HR_UP"]],gene.list)                    +
labs(title="PLASARI_TGFB1_TARGETS_10HR_UP")

#Top pathways DOWN

#Plot14
plotEnrichment(all.gene.sets[["ICHIBA_GRAFT_VERSUS_HOST_DISEASE_D7_UP"]],gene.list)      +
labs(title="ICHIBA_GRAFT_VERSUS_HOST_DISEASE_D7_UP")
#Plot15
plotEnrichment(all.gene.sets[["ICHIBA_GRAFT_VERSUS_HOST_DISEASE_35D_UP"]],gene.list)     +
labs(title="ICHIBA_GRAFT_VERSUS_HOST_DISEASE_35D_UP")
#Plot16
plotEnrichment(all.gene.sets[["ZEMEK_IMMUNE_CHECKPOINT_BLOCKADE_OVARIAN_CANCE
R_OVERLAP_UP"]],gene.list)                                                                    +
labs(title="ZEMEK_IMMUNE_CHECKPOINT_BLOCKADE_OVARIAN_CANCER_OVERLAP_UP"
)

##################################################################################################
#######

#Clustering

load("C:\\Users\\Sophiya John\\Desktop\\J_STATISTICS\\J_STATISTICAL GENOMICS BIOS
799\\DatCluster.RData")
dim(dat)
#[1]    499 118209
# Note: Clinical data : columns 1: 75
#                    Molecular data: column 76 onwards
dim(dat.expr)  # Top 1000 most associated genes with survival

#mydata

rnaseq <- read.csv("C:\\Users\\Sophiya John\\Desktop\\J_STATISTICS\\J_STATISTICAL GENOMICS
BIOS 799\\RNASeq2.txt", header=T, sep="\t")
#rnaseq
names(rnaseq)
#Choose only saline TMG
df <- rnaseq[ -c(3:11) ]
df2 <- df[ -c(8:10) ]
names(df2)
```

```
df2<-as.data.frame(df2)
df2
#Change col names
colnames(df2) <- c('GENES','TMG1','TMG2','TMG3','SALINE1','SALINE2','SALINE3')
#RELOCATE SALINE TO FIRST 3 COLUMNS
df2
library(dplyr)
TMG2 <- df2 %>%
  relocate(SALINE1, .before = TMG1)%>%
  relocate(SALINE2, .before = TMG1)%>%
  relocate(SALINE3, .before = TMG1)

TMG2

#TO SEPARATE ENSEMBL ID AND GENE NAME

library(dplyr)
library(tidyr)

TMG1<-TMG2 %>% separate(GENES, c('ENSEMBLID', 'GENENAME'))
TMG1
TMG<-as.data.frame(TMG1)
TMG<-TMG[,-2]
# Assign the 'ID' column as row names
rownames(TMG) <- TMG$ENSEMBLID

# Remove the 'ID' column from the data frame (if needed)
TMG$ENSEMBLID <- NULL
TMG
dim(TMG)
#[1] 55298     6

dat
dim(dat)
#[1]    499 118210

TMG <- TMG[rowSums(TMG)!=0,]     ## Remove if gene counts is zero for all samples
dim(TMG)
#[1] 25705     6
```

```
### MDS (Multidimensional Scaling) plot:
plotMDS(TMG, main="Multipledimensional scaling plot (MDS)")


keep1 <- rowSums(cpm(as.matrix(TMG))>1)>=2
# At least 2 samples have to have cpm > 1.
data.filtered1 <- TMG[keep1,]
dim(data.filtered1)
#[1] 12775     6
rm(keep1)
dim(data.filtered1)

#d1 <- DGEList(counts=as.matrix(data.filtered1), lib.size=colSums(data.filtered1),
group=c(rep("SALINE",3),rep("TMG",3)))
dim(d1)
d1
d1 <- calcNormFactors(d1, method="TMM")       ## Calculates normalization factors
d1
d1 <- estimateDisp(d1)                        ## Calculates genewise dispersion parameter adjusted using
bayesian empirical method(Recommended, edgeR)
d1



## BCV
plotBCV(d)
plotBCV(d1)

de.test <- exactTest(d,pair=c("Normal","Tumor")) ## First entry under pair is the baseline
de.test.FDR <- topTags(de.test,n=Inf,adjust.method="BH", sort.by="PValue")
head(de.test.FDR$table)
summary(de <- decideTestsDGE(de.test, p=0.05, adjust="BH"))  ## Counts up and down regulated genes

de.test1 <- exactTest(d1,pair=c("SALINE","TMG")) ## First entry under pair is the baseline
de.test1.FDR <- topTags(de.test1,n=Inf,adjust.method="BH", sort.by="PValue")
head(de.test1.FDR$table)
summary(de1 <- decideTestsDGE(de.test1, p=0.05, adjust="BH"))  ## Counts up and down regulated
genes
dim(d1)
```

```
## 1. Hierarchical clustering method
##-----------------------------------------------------------------------------------------------
dd <- as.matrix(dat.expr)

dd <- as.matrix(data.filtered1)
dd <- as.matrix(d1)

require(graphics)
d <- dist(dd, method = "euclidean") # distance matrix
h.clust <- hclust(d, method="complete")
str(h.clust)

plot(h.clust,labels = F,xlab="",sub="") # display dendogram
cluster.mem <- cutree(h.clust, k=3) # cut tree into 3 clusters
# draw dendogram with red borders around the 3 clusters
rect.hclust(h.clust, k=3, border="red")
table(cluster.mem)

plot(h.clust,labels = F,xlab="",sub="") # display dendogram
cluster.mem <- cutree(h.clust, k=2) # cut tree into 3 clusters
# draw dendogram with red borders around the 3 clusters
rect.hclust(h.clust, k=2, border="red")
table(cluster.mem)

plot(h.clust,labels = F,xlab="",sub="") # display dendogram
cluster.mem <- cutree(h.clust, k=5) # cut tree into 2 clusters
# draw dendogram with red borders around the 2 clusters
rect.hclust(h.clust, k=5, border="red")
table(cluster.mem)

plot(h.clust,labels = F,xlab="",sub="") # display dendogram
cluster.mem <- cutree(h.clust, k=10) # cut tree into 2 clusters
# draw dendogram with red borders around the 2 clusters
rect.hclust(h.clust, k=10, border="red")
table(cluster.mem)

plot(h.clust,labels = F,xlab="",sub="") # display dendogram
cluster.mem <- cutree(h.clust, k=20) # cut tree into 2 clusters
# draw dendogram with red borders around the 2 clusters
```

```
rect.hclust(h.clust, k=20, border="red")
table(cluster.mem)

plot(h.clust,labels = F,xlab="",sub="") # display dendogram
cluster.mem <- cutree(h.clust, k=50) # cut tree into 2 clusters
# draw dendogram with red borders around the 2 clusters
rect.hclust(h.clust, k=50, border="red")
table(cluster.mem)

plot(h.clust,labels = F,xlab="",sub="") # display dendogram
cluster.mem <- cutree(h.clust, k=100) # cut tree into 2 clusters
# draw dendogram with red borders around the 2 clusters
rect.hclust(h.clust, k=100, border="red")
table(cluster.mem)

#Not doing this bcoz not clinical data

### Assess the clusters using CoxPH analysis among the clusters
all(names(cluster.mem)==dat$bcr_patient_barcode)
# [1] TRUE
D <- cbind(cluster.mem,dat[,1:75]) # Combine the cluster.id with clinical part of data
D[1:5,1:5]


all(names(cluster.mem)==data.filtered1$rowname)
# [1] TRUE
D <- cbind(cluster.mem,data.filtered1) # Combine the cluster.id with clinical part of data
D[1:5,1:5]


### Log rank test:
###-----------------------------------------------------------------------------------------
library(survival)
lr.test <- survdiff(Surv(time.ttr, recurrence == "YES")~as.factor(cluster.mem), data = D,
na.action=na.exclude)
lr.test
pval <- pchisq(lr.test$chisq,1, lower.tail=F)
pval
### Kaplan Meier plot of the two clusters
for (i in 1:length(unique(sort(cluster.mem))))
```

```
{
  ii <- unique(sort(cluster.mem))[i]
  ddd <- D[D$cluster.mem==ii,]
  f <- survfit(Surv(time.ttr, recurrence == "YES")~1, data = ddd)
  if (i == 1){plot(f,col="black",conf.int=FALSE,xlab="Time to
Recurrence",ylab="Survival",main="",lwd=2)
    clust.name <- "Cluster 1"
    clr <- i}
  else {lines(f,col=i,conf.int=F,lwd=2)
    clust.name <- c(clust.name,paste("Cluster",i))
    clr <- c(clr,i)}
  text(1500,1,paste("p-value = ",format(pval, digits=2,scientific = T)))
}
legend("topright", clust.name, text.col = clr, cex=0.9)

##########################################################################################


## 2. k-means clustering method
##-----------------------------------


#rm(list=setdiff(ls(),c("dat.expr","dat")))
#ls()
#dd <- as.matrix(dat.expr)

#dd <- as.matrix(data.filtered1)

dim(dd)
dd[1:5,1:5]
## A plot of the within groups sum of squares by number of clusters extracted can help determine
## the appropriate number of clusters.
wss <- (nrow(dd)-1)*sum(apply(dd,2,var))
for (i in 2:8) wss[i] <- sum(kmeans(dd, centers=i)$withinss)
plot(1:8, wss, type="b", xlab="Number of Clusters",ylab="Within groups sum of squares")

k.clust <- kmeans(dd,3)
str(k.clust)
cluster.mem <- k.clust$cluster
cluster.mem[1:5]
```

```
table(cluster.mem)

k.clust <- kmeans(dd,2)
str(k.clust)
cluster.mem <- k.clust$cluster
cluster.mem[1:5]
table(cluster.mem)

k.clust <- kmeans(dd,5)
str(k.clust)
cluster.mem <- k.clust$cluster
cluster.mem[1:5]
table(cluster.mem)

k.clust <- kmeans(dd,8)
str(k.clust)
cluster.mem <- k.clust$cluster
cluster.mem[1:5]
table(cluster.mem)

k.clust <- kmeans(dd,10)
str(k.clust)
cluster.mem <- k.clust$cluster
cluster.mem[1:5]
table(cluster.mem)

k.clust <- kmeans(dd,12)
str(k.clust)
cluster.mem <- k.clust$cluster
cluster.mem[1:5]
table(cluster.mem)

### Assess the clusters using CoxPH analysis among the clusters
###
all(names(cluster.mem)==dat$bcr_patient_barcode)
# [1] TRUE
D <- cbind(cluster.mem,dat[,1:75]) # Combine the cluster.id with clinical part of data
D[1:5,1:5]

all(names(cluster.mem)==data.filtered1$rowname)
```

```
# [1] TRUE
D2 <- cbind(cluster.mem,data.filtered1) # Combine the cluster.id with clinical part of data
D2[1:5,1:5]



### Log rank test:
###-----------------------------------------------------------------------------------------
library(survival)
lr.test <- survdiff(Surv(time.ttr, recurrence == "YES")~as.factor(cluster.mem), data = D,
na.action=na.exclude)
lr.test
pval <- pchisq(lr.test$chisq,1, lower.tail=F)
pval
for (i in 1:length(unique(sort(cluster.mem))))
{
  ii <- unique(sort(cluster.mem))[i]
  ddd <- D[D$cluster.mem==ii,]
  f <- survfit(Surv(time.ttr, recurrence == "YES")~1, data = ddd)
  if (i == 1){plot(f,col="black",conf.int=FALSE,xlab="Time to
Recurrence",ylab="Survival",main="",lwd=2)
    clust.name <- "Cluster 1"
    clr <- i}
  else {lines(f,col=i,conf.int=F,lwd=2)
    clust.name <- c(clust.name,paste("Cluster",i))
    clr <- c(clr,i)}
  text(1300,1,paste("p-value = ",format(pval, digits=2,scientific = T)))
}
legend("topright", clust.name, text.col = clr, cex=0.9)

#heatmap
dd
heatmap(dd, main = "SALINE Vs TMG", xlab = "Samples", ylab = "DEG")
#https://www.biostars.org/p/374551/

##########################################################################################
##########
```